

Reviewer Pack — Plethysm Coefficient Ratio in SOS Refutations for Tseitin Fo...

Entry e8c1cae80bbb · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Plethysm Coefficient Ratio in SOS Refutations for Tseitin Formulas

Entry ID: e8c1cae80bbb **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-19 13:22:49 UTC

1. Conjecture

Field A (mathematical branch): Schur-Weyl Duality **Field B** (complexity object): Sum-of-Squares Refutation Size

Statement:

For a Tseitin formula over an n -vertex expander graph, the ratio of plethysm coefficients $\lambda[\mu]$ in the decomposition of $\text{Sym}^{k(\text{Sym}_m(C^n))}$ satisfies $\lambda[\mu] \geq \Omega(n^{\{k/2\}})$ for all $k \leq \log n$, while SOS refutation size is $\Theta(n^{\{k/2\}})$

Rationale (proposer's reasoning):

Schur-Weyl duality provides a combinatorial framework to analyze tensor rank, which directly relates to SOS hierarchy's ability to refute CSPs. The plethysm coefficients capture symmetry-breaking patterns that could create inherent barriers for sos algorithms.

Taxonomy category: SOS_HIERARCHY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): bf442bfce9951161

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 30
    k = random.randint(5, 40)
    m = random.randint(1, 2)

    # Generate a random expander graph (simplified for testing)
    G = {i: set(random.sample(range(n), n // 2)) for i in range(n)}

    # Compute plethysm coefficients using Young tableaux
    def plethysm_coefficient(k, m):
        if k == 0 or m == 0:
            return 1
        coeff = 0
        for i in range(min(k, m) + 1):
            coeff += math.comb(k, i) * math.comb(m, i)
        return coeff

     $\lambda_\mu$  = plethysm_coefficient(k, m)

    # Compute SOS refutation size (simplified for testing)
    sos_refutation_size = n ** (k / 2)

    # Check the conjecture
    if  $\lambda_\mu$  < n ** (k / 2):
        return {
            "metric_name": "plethysm_coefficient_ratio",
            "metric_value":  $\lambda_\mu$ ,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": f"plethysm_coefficient < n^(k/2) for k={k}, m={m}"
        }
    else:
        return {
            "metric_name": "plethysm_coefficient_ratio",
            "metric_value":  $\lambda_\mu$ ,
```

```

        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{result['metric_name']}\", \"metric_value\": {result['metric_value']}\"}}")
        results.append(result)

    mean_metric = sum(r['metric_value'] for r in results) / len(results)
    std_metric = math.sqrt(sum((r['metric_value'] - mean_metric) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r['conjecture_holds']) / len(results)

    if all(r['conjecture_holds'] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result['conjecture_holds'])
        print(f"RESULT: FALSIFIED counterexample=\"plethysm_coefficient < n^(k/2)\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

_coefficient < n^(k/2) for k=18, m=2"}
TRIAL: {"seed": 503, "metric_name": "plethysm_coefficient_ratio", "metric_value": 16, "instances_tested": 1}
TRIAL: {"seed": 547, "metric_name": "plethysm_coefficient_ratio", "metric_value": 8, "instances_tested": 1}
TRIAL: {"seed": 593, "metric_name": "plethysm_coefficient_ratio", "metric_value": 351, "instances_tested": 1}
TRIAL: {"seed": 631, "metric_name": "plethysm_coefficient_ratio", "metric_value": 300, "instances_tested": 1}
TRIAL: {"seed": 677, "metric_name": "plethysm_coefficient_ratio", "metric_value": 10, "instances_tested": 1}
TRIAL: {"seed": 727, "metric_name": "plethysm_coefficient_ratio", "metric_value": 39, "instances_tested": 1}
TRIAL: {"seed": 773, "metric_name": "plethysm_coefficient_ratio", "metric_value": 153, "instances_tested": 1}
TRIAL: {"seed": 821, "metric_name": "plethysm_coefficient_ratio", "metric_value": 703, "instances_tested": 1}
TRIAL: {"seed": 877, "metric_name": "plethysm_coefficient_ratio", "metric_value": 210, "instances_tested": 1}
TRIAL: {"seed": 929, "metric_name": "plethysm_coefficient_ratio", "metric_value": 18, "instances_tested": 1}

```

8. Critic adversarial review

Critic verdict: CHALLENGE

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: Multiple trials show plethysm_coefficient < n^(k/2) for specific k,m parameters, directly contradicting the conjecture | next:

Analyze the $k=7, m=1$ case to understand why plethysm coefficients fail to meet the $\Omega(n^{\{k/2\}})$ bound

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	38321
2	preregistration	ollama_remote	qwen3:8b	0	0	27664
3	novelty	ollama_remote	qwen3:8b	0	0	24089
4	novelty	ollama_remote	qwen3:8b	0	0	23162
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10270
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11784
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7885
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8927
9	critic	ollama_remote	qwen3:8b	0	0	33403
10	judge	ollama_remote	qwen3:8b	0	0	23685

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 209190 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/e8c1cae80bbb.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e8c1cae80bbb.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/e8c1cae80bbb.tar.gz` (if generated)